AP BIOLOGY TOPIC 2.5

Membrane Transport: Passive, Active & Vesicular Bulk Transport

UNIT 2: CELL STRUCTURE AND FUNCTION (10–13% EXAM WEIGHT)

ENE-2.E | ENE-2.F | ELECTROCHEMICAL GRADIENTS | ENDO / EXOCYTOSIS

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§1. KEY TERMS & FORMAL DEFINITIONS LEXICON — TOPIC 2.5 (MEMBRANE TRANSPORT: PASSIVE, ACTIVE & VESICULAR BULK TRANSPORT)

CED Core Vocabulary

Concentration Gradient — Spatial difference in solute concentration across a membrane: Down the Gradient ([High] → [Low]): Spontaneous ($\Delta G < 0$, increases entropy) → Passive...

2. Electrical Gradient (Membrane Potential) — Difference in electrical charge (voltage) across the membrane (cytoplasm is typically negative, −70 mV , relative to extracellular fluid).

Core Requirements — Moves specific solutes against their electrochemical gradient ([Low] → [High]) ; always requires integral carrier/pump proteins AND metabolic energy (ATP) .

1. Endocytosis (Bulk Uptake INTO Cell) — Plasma membrane invaginates (folds inward) to form a pocket that pinches off as an intracellular vesicle: Phagocytosis ("Cell Eating"): Pseudopodia engulf large...

3 Types of Driving Gradients Across Membranes — 1. Chemical Gradient: Difference in solute concentration across the membrane.

3. Electrochemical Gradient — Combined force of both the chemical concentration gradient AND electrical charge gradient acting on a charged ion (Na^+, K^+, H^+, Ca^{2+}, Cl^-).

Primary Active Transport (Direct ATP Coupling) — Hydrolysis of ATP → ADP + P_i directly phosphorylates the transport pump, triggering a conformational shape change that translocates ions uphill.

Pinocytosis ("Cell Drinking") — Non-specific uptake of tiny droplets of extracellular fluid and dissolved solutes into small pinocytotic vesicles.

§2. CONCENTRATION & ELECTROCHEMICAL GRADIENTS

Potential Energy

- **Concentration Gradient:** Spatial difference in solute concentration across a membrane:
  - **Down the Gradient ([High] → [Low]):** Spontaneous ($\Delta G < 0$, increases entropy) → **Passive Transport (NO ATP required)**; leads toward *dynamic equilibrium* (net movement = 0, though molecules keep moving equally in both directions!).
  - **Against the Gradient ([Low] → [High]):** Non-spontaneous ($\Delta G > 0$) → **Active Transport (Requires metabolic energy / ATP)**.
- **3 Types of Driving Gradients Across Membranes:**
  - **1. Chemical Gradient:** Difference in *solute concentration* across the membrane.
  - **2. Electrical Gradient (Membrane Potential):** Difference in *electrical charge (voltage)* across the membrane (cytoplasm is typically negative, **−70 mV** , relative to extracellular fluid).
  - **3. Electrochemical Gradient:** Combined force of both the **chemical concentration gradient AND electrical charge gradient** acting on a charged ion (**Na^+, K^+, H^+, Ca^{2+}, Cl^-**).

[CORE AXIOM / REMEMBER] **A concentration or electrochemical gradient = stored potential energy** (like water behind a dam) that cells can harvest to drive work or ATP synthesis!

[AP EXAM FRQ SCORING STRATEGY] Whenever an AP Biology question asks what drives the movement of **charged ions** (**Na^+, K^+, H^+, Ca^{2+}**) across a membrane, always write **"electrochemical gradient"** (not just concentration gradient)!

§3. MASTER COMPARISON OF MEMBRANE TRANSPORT MECHANISMS

High-Yield Summary Table

TRANSPORT TYPE	ATP ENERGY NEEDED?	DIRECTION (GRADIENT)	MEMBRANE PROTEIN / MECHANISM	CLASSIC EXAMPLES
Simple Diffusion	**No** (Passive)	**High → Low** (Down)	**No** (Directly through phospholipid bilayer)	**O_2, CO_2, N_2**, steroid hormones
Facilitated Diffusion	**No** (Passive)	**High → Low** (Down)	**Yes** (Channel or Carrier proteins)	**H_2O** (aquaporins), Glucose (GLUT), **Na^+/K^+** channels
Primary Active Transport	**Yes** (Direct ATP)	**Low → High** (Against)	**Yes** (ATPase Pump protein changes shape)	**Na^+/K^+** pump (**3 Na^+** out / **2 K^+** in), **H^+** proton pump
Bulk Transport (Endo / Exocytosis)	**Yes** (Requires ATP + cytoskeleton)	Into or Out of cell (Macromolecules)	**Membrane Vesicles** bud or fuse with bilayer	Secretion of insulin/neurotransmitters; WBC phagocytosis

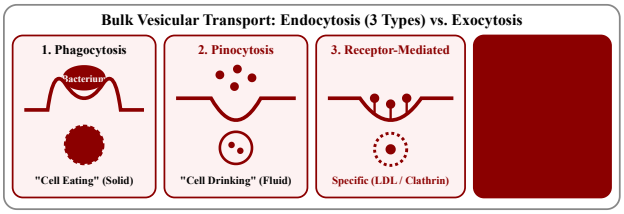
§5. BULK TRANSPORT: ENDOCYTOSIS & EXOCYTOSIS

Macromolecules & Vesicles

HIERARCHICAL CONCEPT & PATHWAY MAP — **[CAUSE ➔ MECHANISM ➔ OUTCOME]**

Bulk Transport ➔ **ATP-requiring membrane vesicle remodeling** ➔
1. Endocytosis (Bulk Uptake INTO Cell) ➔
2. Exocytosis (Bulk Secretion OUT OF Cell)

- Large macromolecules (proteins, polysaccharides, polynucleotides) and whole cells/particles cannot pass through transport proteins → cross via **ATP-requiring membrane vesicle remodeling** (enabled by phospholipid bilayer fluidity):
- **1. Endocytosis (Bulk Uptake INTO Cell):** Plasma membrane invaginates (folds inward) to form a pocket that pinches off as an intracellular vesicle:
  - **Phagocytosis ("Cell Eating"):** Pseudopodia engulf large solid particles, cell debris, or whole bacteria into a **food vacuole (phagosome)**, which fuses with a **lysosome** for digestion (e.g., *Amoeba* feeding; human macrophages engulfing pathogens).
  - **Pinocytosis ("Cell Drinking"):** Non-specific uptake of tiny droplets of extracellular fluid and dissolved solutes into small pinocytotic vesicles.
  - **Receptor-Mediated Endocytosis (Highly Specific):** Extracellular ligands bind specific membrane **receptor proteins** clustered in **clathrin-coated pits** → folds inward to form a coated vesicle (e.g., human liver cells taking up **cholesterol-carrying LDL particles**; defective in *familial hypercholesterolemia*!).
- **2. Exocytosis (Bulk Secretion OUT OF Cell):**
  - Intracellular transport/secretory vesicles (budded from the **trans-Golgi**) migrate along **microtubule** tracks to the plasma membrane.
  - Vesicle lipid bilayer **fuses with the plasma membrane**, expelling its contents outside while adding new phospholipids & proteins to expand the plasma membrane!
  - **Key Biological Roles:** Secretion of peptide **hormones** (insulin from pancreatic β-cells), **neurotransmitters** (acetylcholine/dopamine at synaptic clefts), digestive enzymes, antibodies, plant cell wall polysaccharides, and waste expulsion.



§4. ACTIVE TRANSPORT: MOVING UPHILL AGAINST THE GRADIENT

Requires ATP

- **Core Requirements:** Moves specific solutes **against their electrochemical gradient (** **[Low] → [High]** **)**; always requires **integral carrier/pump proteins** AND **metabolic energy (ATP)**.
- **Primary Active Transport (Direct ATP Coupling):**
  - Hydrolysis of **$ATP \rightarrow ADP + P_i$** directly phosphorylates the transport pump, triggering a **conformational shape change** that translocates ions uphill.
  - **Classic Example — Na^+/K^+ ATPase Pump:** Pumps **3 Na^+** **OUT** of the cell and **2 K^+** **INTO** the cell per 1 ATP hydrolyzed, maintaining a negative resting potential inside.

§6. WHY MEMBRANE TRANSPORT MATTERS FOR LIFE

Cellular Homeostasis

- **Homeostasis & Bioenergetics:** Cells cannot exist at chemical equilibrium with their surroundings (equilibrium = death!). Coordinated transport maintains:
  - **Resting & Action Potentials** across neuron and muscle membranes.
  - **Proton Gradients (** **H^+** **)** for oxidative phosphorylation and photophosphorylation.
  - **Selective Nutrient Import** (glucose, amino acids) & **Metabolic Waste Export (** **CO_2** **, urea, lactate).**
  - **Intercellular Communication** via exocytosis of neurotransmitters and hormones.

[CORE AXIOM / REMEMBER] **Cells constantly balance two complementary strategies:**

- **Passive Transport (Simple + Facilitated):** Energy-efficient (**0 ATP**), rapid movement *down* existing gradients.
- **Active + Bulk Transport:** Energy-investing (**uses ATP**), tightly regulated control *against* gradients or for large cargo.

§7. SCHOLARLY SYNTHESIS: EXPERIMENTAL DESIGN, QUANTITATIVE ANALYSIS, PATHOLOGY & AP FRQ RHETORIC

High-Yield Exam Rigor

MASTER TOPIC SYNTHESIS & CAUSAL ARCHITECTURE — TOPIC 2.5 **[MOLECULAR CAUSE ➔ CELLULAR MECHANISM ➔ PHYSIOLOGICAL / ECOLOGICAL EFFECT ➔ FRQ JUSTIFICATION]**

Topic 2.5: Membrane Transport: Passive, Active & Vesicular Bulk Transport ➔ **Structural / Energetic Determinants** ➔ **Regulated Pathway & Homeostasis** ➔ **Experimental Perturbation (±2 SE_x̄)** ➔ **Phenotypic / Evolutionary Outcome**

SCHOLARLY DIMENSION	COMPREHENSIVE ACADEMIC REFERENCE & MECHANISTIC SPECIFICATION
I. Experimental Design & Required Lab Controls	AP Investigation 4 (Diffusion & Osmosis): Dialysis tubing & potato core mass change (% ΔMass = [(Final − Initial) / Initial] × 100) across sucrose molarities (0.0 M to 1.0 M). IV: Extracellular sucrose molarity (M); DV: % change in mass; Isotonic point: x-intercept where % ΔMass = 0% (no net water movement). **Null Hypothesis (H_0):** State explicitly that the independent variable treatment has *no statistically significant effect* on the dependent variable; any observed difference is due to random sampling chance alone.
II. Quantitative Formulas & Graph Interpretation	Quantitative Formulas & Graph Rules: Total Water Potential: $\Psi = \Psi_p + \Psi_s$. Solute Potential: $\Psi_s = -iCRT$ (where i = ionization constant [1 for sucrose/glucose, 2 for NaCl], C = molarity in mol/L, R = 0.0831 L·bar/mol·K, T = K = °C + 273). Surface-Area-to-Volume Ratio (SA:V): Cube SA = $6s^2$, V = s^3 (SA:V = 6/s). Larger SA:V accelerates diffusion per unit volume. **Error Bar Rule (±2 SE_x̄):** If the ±2 SE_x̄ error bars of two treatment groups *do not overlap*, there is a statistically significant difference between means ($p < 0.05$; reject H_0).
III. Pathology, Mutation & Disruption Case Study	Clinical & Physiological Disruption Case Study: Cystic Fibrosis results from a deletion (ΔF508) in the CFTR chloride channel protein, preventing misfolded channels from reaching the apical plasma membrane. Impaired Cl^- efflux prevents osmotic water movement into mucus, producing thick, dehydrated mucus in lungs and pancreatic ducts. Ouabain/Digoxin inhibits the Na^+/K^+-ATPase pump, collapsing the electrochemical gradient required for secondary active cotransport.
IV. AP FRQ Task Verb Scoring Templates	• **DESCRIBE / IDENTIFY:** State the direct trend, value, or structural component with exact units from the figure. • **EXPLAIN (How/Why):** Connect cause to effect via a 3-link chain: *"[Change in X] causes [molecular/cellular mechanism Y], which results in [observable outcome Z]."* • **PREDICT:** State the specific directional change (increase, decrease, or remain constant) when the pathway is perturbed. • **JUSTIFY:** Pair quantitative data/evidence from the prompt with the underlying biological principle from Topic 2.5 to prove why the claim holds.

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